Fusion of Spectral–Spatial Classifiers for Hyperspectral Image Classification

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***Abstract*— A spectral–spatial (SS) hyperspectral classifier generally implements a spectral classifier (SC) followed by a spatial filter (SF) for classification. This article develops a new approach to fusing multiple SC-SF classifiers for hyperspectral image classification (HSIC) as to improve classification performance. To accomplish this goal an iterative process is particularly designed to fuse the spatial-filtered classification maps (SFMaps) produced by each of SC-SF classifiers into one single SFMap via maximum *a posteriori* (MAP) criterion. Such fused SFMaps are then fed back and added to the current data cube to create a new data set for next round SC-SF classifier fusion. The same process is repeated iteratively until it satisfies an automatic stopping rule. To further fuse more than two SS methods, two approaches are also developed, called simultaneous multiple SC-SF fusion (SMSSF) method and progressive multiple SC-SF fusion (PMSSF) method. Experimental results demonstrate that fusing multiple SC-SF classifiers can indeed perform better than using an individual single SC-SF classifier alone without fusion.**

***Index Terms*— Constrained energy minimization (CEM), edge-preserving filtering (EPF), hyperspectral image classification (HSIC), progressive multiple SS fusion (PMSSF), simultaneous multiple SC-SF fusion (SMSSF), spatial filtered classification map (SFMap), spectral classifier-spatial filter (SC-SF), spectral–spatial (SS).**

NOMENCLATURE

AA	Average accuracy.
APR	Average PR.
AVIRIS	Airborne Visible/Infrared Imaging Spectrometer.
BKG	Background.
CEM	Constrained energy minimization.
EPF	Edge-preserving filter.
HSI	Hyperspectral image.
HSIC	Hyperspectral image classification.
ICEM	Iterative CEM.
ISVM	Iterative support vector machine.
MAP	Maximum *a posteriori*.
OA	Overall accuracy.
OPR	Overall precision.
OTVCA	Orthogonal total variation component analysis.
PMSSF	Progressive multiple SS fusion.
PR	Precision rate.
SC	Spectral classifier/classification.
SF	Spatial filter.
SFMaps	Spatial-filtered classification maps.
SMSSF	Simultaneous multiple SC-SF fusion.
SS	spectral–spatial.
SVM	Support vector machine.
TI	Tanimoto index.

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I. INTRODUCTION

HSIC generally requires both spectral and spatial information to perform well. Over the past decades many research efforts have been focused on HSIC, just name a few [1]–[22], specifically SS methods such as [16]–[22], which have been widely used and proven to be very effective in HSIC. The reason why a hyperspectral image is called "hyperspectral" is because its spectral properties are vital and crucial in data analysis. Accordingly, to perform HSIC effectively, preserving hyperspectral features is more important than spatial properties. However, unlike applications in subpixel target detection, endmember extraction, spectral unmixing which can be primarily characterized by spectral properties, classification generally requires additional spatial information surrounding spectral-classified data sample vectors. Consequently, despite that many approaches to HSIC have been developed in the past SS methods remain one of the most popular and widely used classification methods. This is mainly due to the use of hundreds of contiguous spectral bands provided by a hyperspectral imaging sensor to obtain rich spectral information to accomplish what traditional spatial domain-based image processing and multispectral imaging techniques cannot. This is particularly true for HSIC where small classes can be better characterized by their spectral properties rather than their spatial properties since their spatial information may be limited to

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their small sample pools. Under such circumstances capturing spectral information is critical because any spatial process may sacrifice or compromise spectral characteristics if it is implemented prior to spectral processing. SS methods arise from this need and are specifically designed to first extract spectral information by an SC followed by an SF to capture spatial information from the SC classification map. Owing to confusion with SS methods, the SS method to be discussed in this article is better specified by an SC-SF method where the first "SC" is a generic term to indicate an SC to be used for spectral classification and the second "SF" is also a generic term to specify an SF to be used to extract spatial information. Nevertheless, whenever no confusion occurs, SS and SC-SF will be used interchangeably throughout this article.

One of the most widely used SC methods is SVM. Unfortunately, SVM does not take into account the spatial correlation among pixels. Therefore, SFs are generally required to be used as a follow-up process to capture spatial information among SVM-classified data samples. As a result, many SS methods have been developed by including an SC in conjunction with an SF to extract pixel spatial information to improve classification [1], [4], [10]–[22]. Of particular interest is the approach developed in [11], called EPF-based methods, which used SVM to perform spectral classification followed by spatial domain-based techniques to capture spatial contextual information from SVM-classification maps. Such spatial information obtained directly from SVM-classified data samples is referred to as *a posteriori* spatial information in this article. Another interesting approach was developed in [17], called composite kernel (CK)-based method, which utilized spatial windows to extract spatial information directly from the original hyperspectral data to capture both spectral and spatial information. The spatial information obtained directly from data samples within windows prior to SVM classification is referred to as *a priori* spatial information in this article. Unfortunately, such CK-based approach can be easily affected by noise and wide variations in grayscale within classes.

By realizing the benefits of fusing multiple SS classifiers this article investigates a fusion problem of classifiers with a particular structure, namely, SC-SF classifiers. Specifically, it develops an approach to fusing different SC-SF classifiers with SCs characterized by two specific SCs, pure pixel-based classifiers and mixed pixel-based classifiers to perform spectral classification and various types of SFs to extract spatial information so as to improve classification performance. To fuse SS information produced by each such SC-SF classifier, an iterative process is developed to yield fused SS maps which are further fed back to augment the current data cube to create a new data cube as a new image for reimplementing SC-SF classifiers. The same process is repeated by adding more SS information in an iterative manner until a stopping rule is satisfied in which case the fusion process is terminated.

For more than two SC-SF classifiers to be fused, two approaches are further developed. The first one is called SMSSF method which fuses all SC-SF classifiers all together at the same time in one shot operation. Compared to SMSSF a second one is called PMSSF method which fuses any arbitrary pair of two SC-SF classifiers at a time in a progressive manner. Most importantly, PMSSF can fuse any two arbitrary SC-SF classifiers to dictate the progressive changes in classification of each SC-SF classifier to be fused.

Finally, four popular real hyperspectral images are used for experiments to demonstrate applicability of fusing SC-SF classifiers to HSIC.

II. Motivations and Novelties

The motivations of fusing SS classifiers are originated from a need of how to take advantage of strengths resulting from SS classifiers of different types such as pure-pixel based classifiers and mixed pixel classifiers implemented for spectral classification and various SFs used to capture spatial information in SS methods to improve classification performance over SS classifiers used alone. Specifically, in hyperspectral imaging an SC can be performed either at full pixel scale as pure pixel classification such as SVM or at subpixel scale as a mixed pixel classification such as orthogonal subspace projection (OSP) [23]. As for SFs there are also several filters that can be used for spatial processing such as Gaussian filters [24], guided filter [25], Gabor filter [26], EPF [11], and SS feedback close network systems [27]. Combining a different SC with a different SF yields a different classification result. For example, SVM-EPF in [11], referred to as SVM-EPF classifier, has shown to perform better than most existing hyperspectral image classifiers. On the other hand, ICEM coupled with a Gaussian filter in [28], referred to as CEM-Gaussian classifier, has also shown promising in HSIC where CEM is an SC implemented as mixed pixel classifiers. So, an interesting question is "Is there any way to fuse these two heterogeneous classifiers, SVM-EPF classifier and CEM-Gaussian classifier, to improve classification results?" If it is doable, "how can it be done?"

There are many SC-SF classifiers developed in the literature using different types of SCs and SFs. Since each SC-SF classifier has its own merit in classification, it is highly desirable if we can fuse multiple SC-SF classifiers to take advantage of their strengths to improve classification performance. However, this is easily said than done. This article develops an approach with several novelties to address the above issues.

1) The first major novel idea is to fuse multiple SC-SF classifiers by looking into the structure of an SC-SF classifier to develop a three-stage fusion process. At the first stage, an SC such as SVM is used to perform spectral classification to produce discrete spectral classification maps. To capture spatial information among spectral-classified data samples an SF is further used at the second stage to capture spatial information surrounding spectral-classified data samples obtained at the first stage to produce continuous-valued SFMaps. Finally, at the third stage, final discrete-valued classification maps are generated by binarizing the continuous-valued SFMaps generated at the second stage. With this three-stage structure the fusion should occur right after the second stage and before third stage since the results

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from the second stage are continuous-valued SFMaps which retain spatial information before binarization to produce final discrete-valued classification maps in the third stage.

2) Another major novel idea is to design an iterative process to fuse multiple SC-SF classifiers. Its key concept is motivated by and originated from the celebrated expectation-maximization (EM) algorithm [30], [31] which has been widely used in statistical signal processing to estimate parameters from an incomplete data set. The EM algorithm finds the conditional expectation based on observed data samples from an on-going random process and updates the information by maximizing an objective function. More specifically, it begins with prior knowledge provided by incomplete data, referred to as *a priori* knowledge, and then it introduces its own knowledge obtained from observations, referred to as *a posteriori* knowledge, to make data knowledge complete to find unconditional expectation which is the first part of EM algorithm, called expectation step (E-step). If the found expectation is not satisfactory, it indicates that the introduced *a posteriori* knowledge needed to be updated. To accomplish this task, it maximizes an objective cost function and uses the found maximization solution to replace the current *a posteriori* information. This is the second part of the EM algorithm, called maximization step (M-step). The same process is repeated iteratively until a stopping rule is satisfied. Basically, the EM algorithm uses observation obtained *a posteriori* knowledge to update incomplete *a priori* knowledge via an iterative process. The fusion idea proposed in this article takes advantage of the EM algorithm with a slight twist from a different interpretation. It assumes that the original data cube carries incomplete classification information provided by a limited set of training samples as *a priori* knowledge. It first begins with introducing its own *a posteriori* knowledge by adding the SFMaps which are obtained by fusing the continuous-valued SFMaps produced by each of SC-SF classifiers to the current data cube to make incomplete data knowledge complete in the sense of classification. It then calculates classification accuracy such as OA to work as if it performs the E-step in the EM algorithm. This is followed by a fusion which can be interpreted as the M-step in the EM algorithm by performing the MAP derived from detection theory [32], which takes the maximal value of multiple continuous-valued SFMaps for each data sample as the value of its corresponding data sample in the fused SFMaps. If the introduced *a posteriori* classification is still insufficient, then the fused SFMaps are once again fed back to be added to the current data cube to create a new data cube for next round processing of SC-SF classification. Such EM algorithm-like fusion process comprises two operations, performing spatial filtering on the spectral classification maps as the E-step to produce SFmaps and augmenting the current data cube by adding the fused SFMaps as the M-step to produce updated *a posteriori* classification knowledge. It is our belief that using the EM algorithm to design a process of fusing multiple SC-SF classifiers has never been explored in the literature.

3) As noted above, a key idea is how to design a criterion to automatically terminate the iterative process. So, a third novel idea is to develop an automatic stopping rule which uses TI in [33] as a criterion to measure the overlap between classification results obtained by two consecutive iterations. Once the iterative process is completed, the resulting fused continuous SFMap is then input to the third stage for final classification.

4) When more than two SC-SF classifiers are to be fused, an issue is how to effectively fuse multiple classifiers. This is similar to an issue of how to extend a two-class classification to a multiple-class classification. To address this issue a fourth novel idea is to develop two types of fusing multiple SC-SF classifiers. One is SMSSF which is derived from multiple-class classification by interpreting each SC-SF classifier as one class so that extending two SS classifiers to multiple SS classifiers can be done in exactly the same way that binary classification is extended to multiple-class classification. Accordingly, SMSSF can be implemented in two versions according to one-against-all strategy (i.e., one-versus-the rest) and one-against-one strategy in [34]. The other approach to fuse multiple SS classifiers is PMSSF which is quite different from SMSSF in the sense that PMSSF fuses any pair of two SC-SF classifiers at a time progressively instead of fusing multiple SS classifiers all together. It should be noted that the two SC-SF classifiers to be fused by PMSSF can be either a single unfused SC-SF classifier or fused SC-SF classifier. A key difference between SMSSF and PMSSF is that SMSSF is a one-shot operation, while PMSSF provides fusion successive profiles of any two SC-SF classifiers regardless of whether an SC-SF classifier is fused or not.

5) Finally and also most importantly, fusing spectral and spatial features for classification has been studied in the past [35]–[37]. But fusing two different classifiers seems not being investigated. The key difference between fusion of SS features and fusion of SS classifiers lies on the fact that the former is generally referred to as data fusion, which fuses data information regardless of sensors to be used for data acquisition, compared to the latter which is sensor fusion that considers a classifier as a sensor and fuse different types of sensors in their structures. This is similar to the difference between implementing kernels on data and on classifiers respectively.

Of course, there are many other different classifiers studied extensively in the literature such as spatial-spectral classifiers [4], [17], convolutional neural networks (CNNs) [38]–[40], and dynamic graph convolutional network [41]–[42]. Unfortunately, the structures of these classifiers are all different from that of SC-SF classifiers. Since the proposed fusion approach is particularly designed to take advantage of the structures of SC-SF classifiers, it is not directly applicable to these types of classifiers. Nevertheless,

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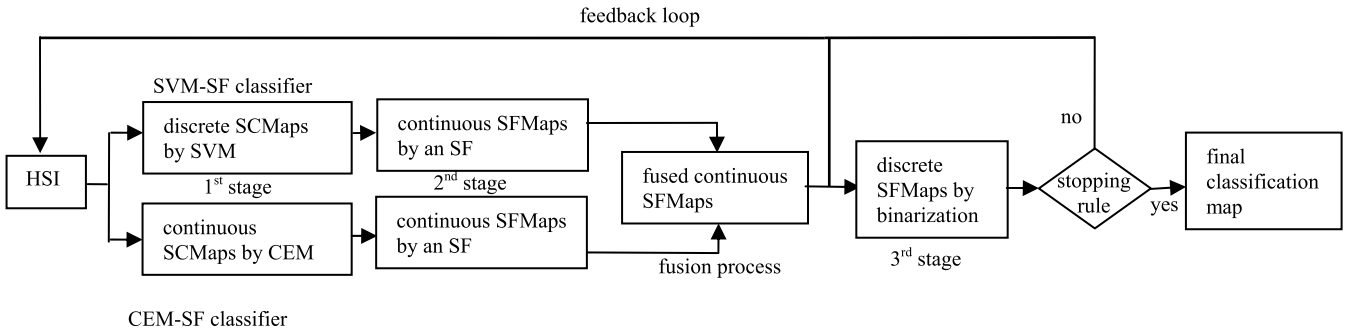


Fig. 1. Block diagram of fusion of SVM-SF and CEM/LCMV-SF classifiers.

their fusion can be considered as a future work and thus not be considered in this article.

III. SS Classifiers

An SC generally does not take care of spatial information surrounding its classified data samples. For example, a widely used SC is SVM which is a pixel-based classifier and does not take into account interpixel spatial information. However, in HSIC, spatial information is crucial for classification. To address this issue, an SC such as SVM should include spatial information among data samples to improve classification. Accordingly, many SC-SF classifiers have been developed by implementing an SC in conjunction with an SF to extract intersample spatial information for better classification [16]–[22]. For example, an SVM-SF classifier first implements SVM as an SC to perform spectral classification, then uses an SF to extract spatial information from SVM-classification maps to improve classification performance. Among SC-SF classifiers one particular interesting approach is the one developed in [11], which implements an SVM classifier as an SC followed by an SF specified by an EPF. Such SVM-EPF methods have been shown to outperform many existing HSIC methods where "SF" is specified by EPF.

In many cases, an SC-SF classifier can be specified by a particular SC or SF to reflect more accurately the nature of an SC-SF classifier. For example, if an SVM classifier is used to perform spectral classification without specifying an SF, then the SC-SF classifier is referred to as SVM-SF classifier. Similarly, if the SF is chosen to be a Gaussian filter without specifying an SC, then the SC-SF classifier is referred to as SC-Gaussian classifier. If both SVM and Gaussian filter are used to specify SC and SF, the resulting SC-SF classifier becomes SVM-Gaussian classifier.

As an alternative, another type of SC-SF classifiers is also considered where the pure pixel-based SVM discussed above is replaced by a subpixel detector as a subpixel-based SC, CEM in [43]–[46]. The resulting SC-SF classifier is referred to as CEM-SF classifier. In particular, the SF in CEM-SF is specified by a Gaussian filter to become a CEM-Gaussian classifier [28].

Since SVM-SF classifiers and CEM-SF classifiers have their own advantages in HSIC and are completely different from each other in terms of their design rationales, it is highly desirable if an SVM-SF classifier can be integrated and fused with CEM-SF classifier to combine their strengths. Unfortunately, a main issue is that SVM is a discrete-valued classifier compared to CEM which is a fractional abundance subpixel detector which is a mixed pixel classifier. How to fuse a pure pixel classifier and mixed pixel classifier is a great challenge. In doing so, we need to explore how an SC-SF classifier is implemented.

As noted in the introduction, an SVM-SF classifier produces three sets of classification maps for each class of interest: 1) discrete-valued spectral classification maps (SCMaps) in the first stage; followed by 2) continuous-valued classification maps resulting from SFs, referred to as SFMaps in the second stage; and concluded by 3) discrete classification maps by binarization in the third stage for final classification. That is, SVM-SF-based classifiers implement a discrete–continuous–discrete value classification in cascade in three stages. By contrast, CEM-SF methods also produce three sets of classification maps in three different stages: 1) continuous-valued class abundance fractional maps, referred to as continuous-valued SCMaps in the first stage derived from CEM; 2) continuous-valued SFMaps by SFs in the second stage; and 3) discrete classification maps by binarization in the third stage to also yield a three-stage continuous–continuous–discrete valued classification in a cascaded chain. By examining their structures, a fusion process should take place and be sandwiched between the second stage and the third stage as shown in Fig. 1. This is because the SCMaps obtained in the first stage only contain spectral information without spatial information. The continuous-valued SFMaps derived from SFs in the second stage contain not only spectral information but also crucial spatial information among spectral classified data samples. So, these two pieces of SFMaps should be integrated and fused to provide better classification information.

The fusion of an SVM-SF classifier and a CEM-SF classifier proposed in Fig. 1 takes advantage of hard decisions made by a pure pixel SC and abundance fractions resulting from a mixed pixel classifier. It fuses the continuous-valued SFMaps produced by the two SC-SF classifiers in the second stage and feeds the fused continuous-valued SFMaps back to the currently being processed data to create a new data cube in an iterative manner. Coincidentally, the proposed fusion shares a similar idea that was used to derive ICEM [28] with a key difference in how to implement iterative processes. Unlike ICEM which implements an iterative process on individual single CEM, the iterative process in Fig. 1 merges two continuous-valued SFMaps of classified data samples to one

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continuous-valued SFMap that can be further binarized in the third stage to produce final discrete-valued classification maps. Such fusion process is carried out by repeatedly feeding back the fused SFMaps iteratively. That is, the more fused SFMaps are fed back, the more interclassified data sample spatial information is fused, thus the better the classification is. If the discretized versions of the two fused continuous-valued SFMaps produced by two consecutive iterations in the third stage do not satisfy a stopping rule, then the last fused continuous-valued SFMaps before binarization are then fed back to be added to the current image data to create a new set of image data for next round fusion. The same process is repeated in an iterative manner until it satisfies the stopping rule.

Most interestingly, a similar idea of the above iterative process carried out in the fusion process in Fig. 1 was used to extend SVM-EPF to an ISVM-EPF as shown in [47].

IV. Fusion of Two SC-SF Classifiers

To illustrate the key idea of fusing multiple SC-SF classifiers, we first present how to fuse two specific SC-SF classifiers in this section and then extend it to fusion of three or more SC-SF classifiers in Section V.

According to Fig. 1, in order to make the fusion of SVM-SF and CEM methods work there are two major steps. One includes a close loop to feed the fused continuous SFMap back to the current data cube so that the fused spectral/spatial information can be updated through feedback loops iteratively. The other determines when the iterative process should be terminated. Its detailed implementation can be described as follows.

For any given pair of two SS classifiers, SS(1) and SS(2), this section develops an approach to fusing SS(1) and SS(2), denoted by SS(1) $\vee$ SS(2). It integrates continuous and discrete classification maps resulting from the SFMaps produced by two SS classifiers. If their discrete classification maps do not meet an automatic stopping rule, their continuous classification maps are then fused by taking the maximal value of their corresponding values to produce a fused classification map which is further fed back to augment the current image data to create a new set of image data for further processing.

Assume that the SC and SF used by SS(j) are represented by SC(j) and SF(j) respectively for $j = 1$ and 2. The main idea of SS(1) $\vee$ SS(2) is similar to that used to derive ICEM [28], both of which use iterative processes to feed spatial information captured by a Gaussian filter back to the original data space. The details of implementing SS(1) $\vee$ SS(2) step-by-step are described as follows.

Since TI measures the overlap between classification maps produced by two consecutive iterations, it may fluctuate as the iterative process goes on. In this case, the stopping rules (2) and (4) specified by step 7 in the SS fusion algorithm described above are particularly designed to terminate the fusion process. Three notes on these stopping rules are worthwhile.

1) If $\{\mathrm{TI}_i^{(k)}\}_k$ is a monotonically increasing function of $\mathrm{TI}_i^{(k)}$ at k bounded above by 1, we can assign a threshold ε in (2) to terminate the fusion process.
2) On the other hand, $\mathrm{TI}_i^{(k)} < \mathrm{TI}_i^{(k-1)}$ may occur for some integer k. In this case, the stopping rule specified by (4) will take care that the monotonical increasing property of $\{\mathrm{TI}_i^{(k)}\}_k$ is violated and the algorithm must be terminated at the kth iteration.

Fusion of SS(1) and SS(2) (SS(1) $\vee$ SS(2))

1) Initial condition: Let $\Omega^{(0)} = \{\mathrm{HSI}\}$ and ε be a prescribed threshold. Set $k = 0$
2) First stage:
   At the kth iteration, implement the spectral classifiers SC(1) and SC(2) on $\Omega^{(k)}$ respectively to produce spectral classification maps of each class, C_i, $1 \le i \le M$, denoted by $\left\{\mathrm{SCMap}(1)_i^{(k)}\right\}_{i=1}^{M}$ and $\left\{\mathrm{SCMap}(2)_i^{(k)}\right\}_{i=1}^{M}$.
3) Second stage:
   Implement the spatial filters SF(1) and SF(2) on $\left\{\mathrm{SCMap}(1)_i^{(k)}\right\}_{i=1}^{M}$ and $\left\{\mathrm{SCMap}(2)_i^{(k)}\right\}_{i=1}^{M}$ respectively to produce spatial filtered maps of each class, C_i, $1 \le i \le M$, denoted by $\left\{\mathrm{SFMap}(1)_i^{(k)}\right\}_{i=1}^{M}$ and $\left\{\mathrm{SFMap}(2)_i^{(k)}\right\}_{i=1}^{M}$.
4) Fusion process:
   For each pixel (x, y) find

$$\text{fused SFMap}_i^{(k)}(x,y) = \max\left\{\mathrm{SFMap}(1)_i^{(k)}(x,y), \mathrm{SFMap}(2)_i^{(k)}(x,y)\right\} \quad (1)$$

5) Third stage: discretization
   Use Otsu's method to threshold each of $\left\{\text{fused SFMap}_i^{(k)}\right\}_{i=1}^{M}$ to produce M binary images for classification maps, one for each class, denoted by $\left\{\mathrm{BMAX}_i^{(k)}\right\}_{i=1}^{M}$.
6) If $k = 0$ or 1, go to step 8. Otherwise, continue.
7) Check TI defined in [33] for two cases
   if

$$\mathrm{TI}_i^{(k)} > \varepsilon \quad (2)$$

   where

$$\mathrm{TI}_i^{(k)} = \frac{|\mathrm{BMAX}_i^{(k)} \cap \mathrm{BMAX}_i^{(k-1)}|}{|\mathrm{BMAX}_i^{(k)} \cup \mathrm{BMAX}_i^{(k-1)}|} \text{ for } 1 \le i \le M \quad (3)$$

   with $|S|$ being the size of set S, and go step 10,

   else $\mathrm{TI}_i^{(k)} < \mathrm{TI}_i^{(k-1)}$ (4)

   If yes, go to step 9. Otherwise, continue.
8) Feedback loop:
   Let $k \leftarrow k + 1$ and form a new data cube

$$\Omega^{(k)} = \Omega^{(k-1)} \cup \left\{\text{fused SFMap}_1^{(k)}\right\} \cup \cdots \cup \left\{\text{fused SFMap}_M^{(k)}\right\} \quad (5)$$

   Go to step 2.
9) SS(1) $\vee$ SS(2) is terminated and $\left\{\mathrm{BMAX}_i^{(k)}\right\}$ are the final classification maps.

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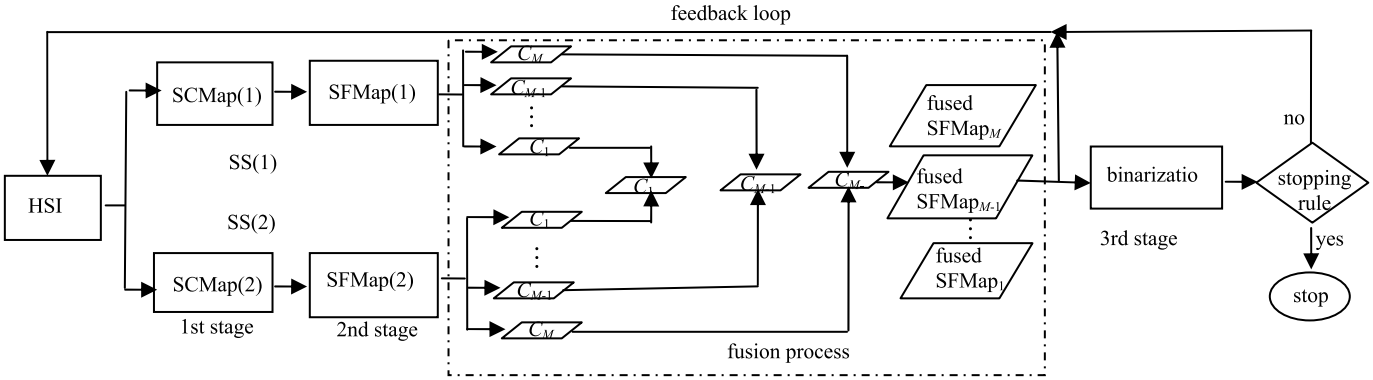


Fig. 2. Diagram of implementing SS(1) ∨ SS(2).

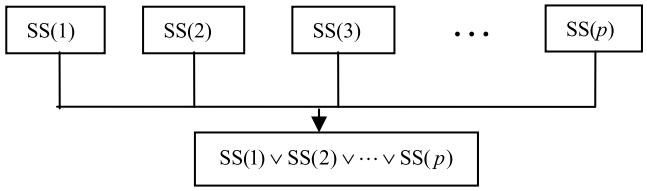


Fig. 3. Diagram of simultaneous fusion of p SS classifiers for SMSSF method.

3) The stopping rule in (4) can be also used to specify a case that when two classifiers already perform very well, there is no need for them to be fused to produce a better fused classifier. That is, when $\mathrm{TI}_i^{(k)} < \mathrm{TI}_i^{(k-1)}$ for $k = 2$, it indicates that the fused classifier does not improve the individual classifiers to be fused in the beginning. So, no fusion process is necessary. Otherwise, $\mathrm{TI}_i^{(k)}$keeps increasing monotonically as k is increased. This implies that the fused classifiers keep improving the individual classifiers to be fused due to its classification improvement by $\mathrm{TI}_i^{(k)} > \mathrm{TI}_i^{(k-1)}$. In this case, we need the stopping rule (2) by setting a threshold ε such as 0.99 to terminate the fusion process.

Fig. 2 depicts a block diagram of implementing SS(1)∨SS(2).

V. Fusion of Multiple SC-SF Classifiers

Suppose that there are p SS (SC-SF) classifiers, $\{\mathrm{SS}(j)\}_{j=1}^{p}$, to be fused. This section extends fusion of two SS classifiers presented in Section IV to fusion of multiple SS classifiers in two different ways.

A. SMSSF Classifier

A general fusion of multiple SS classifiers is called simultaneous fusion shown in Fig. 3 which fuses a set of p SS classifiers, $\{\mathrm{SS}(j)\}_{j=1}^{p}$ simultaneously.

It follows a similar idea of M-class classification [34] where there are two ways to perform classification of M classes. One is to use a strategy of one-against-all policy to perform p-class classification. The other is a strategy of one-against-one policy to perform $C_2^M = M(M-1)/2$ pairs of binary classification.

1) One-Against-Rest Fusion: The one-against-all fusion of p SC-SF classifiers takes the maximal value of continuous spatial-filtered SFMaps in Fig. 1 produced by the j SFs, $\{\mathrm{SF}(j)\}_{j=1}^{p}$. Specifically, the algorithms implementing the fusion of SS(1) and SS(2) (SS(1) ∨ SS(2)) described in Section IV can be easily extended by replacing (1) in step 4 with (4). For each pixel (x, y) find

$$\text{fused SFMap}_i^{(k)}(x,y) = \max_{1\le j\le p}\left\{\mathrm{SFMap}(j)_i^{(k)}(x,y)\right\}. \quad (6)$$

It should be noted that (5) is derived from MAP [30].

2) One-Against-One Fusion: In contrast to the one-against-rest fusion of p SC-SF classifiers described in Section V-A1, the one-against-one fusion of p SC-SF classifiers presented in this subsection fuses all possible pairs of two SC-SF classifiers among p SC-SF classifiers in the same way that the fusion of two SC-SF classifiers developed in Section IV. In this case, there are $C_2^p = p(p-1)/2$ pairs of arbitrarily selected two SC-SF classifiers to be fused. That is

$$\mathbf{r} \in C_{i^*} \text{ if } i^* = \arg\{\max_{1\le i\le M} n_i(\mathbf{r})\} \quad (7)$$

where $n_i(\mathbf{r})$ is the total number of pairs among p SC-SF classifiers claiming the data sample vector $\mathbf{r}$ to be classified in class C_i and $\sum_{i=1}^{M} n_i(\mathbf{r}) = p(p-1)/2$.

B. PMSSF Method

As an alternative to simultaneous fusion of p SC-SF classifiers, we can also develop a fusion process that fuses p SC-SF classifiers in multiple stages with two at a time in one stage in a progressive manner as shown in Fig. 4 where each stage fuses two SS methods, $\mathrm{PSSF}(2) = \mathrm{SS}(1) \vee \mathrm{SS}(2)$ and $\mathrm{PSSF}(j) = \mathrm{PSSF}(j-1) \vee \mathrm{SS}(j)$ with $j > 2$.

It should be noted that Fig. 4 only shows one scenario to perform progressive fusion. As a matter of fact, a general progressive fusion can start out with any arbitrary pair of p SC-SF classifiers in the first stage to produce PSSF(2), then fuses PSSF(2) with any arbitrary SC-SF classifier from the remaining $(p-2)$ SC-SF classifiers in the second stage to yield PSSF(3), and so on. The entire process is repeated $(p-1)$ times. However, it should be noted that the order of implementing PMSSF method in fact has impact on the classification results as will be demonstrated in the experiments conducted in Section VIII.

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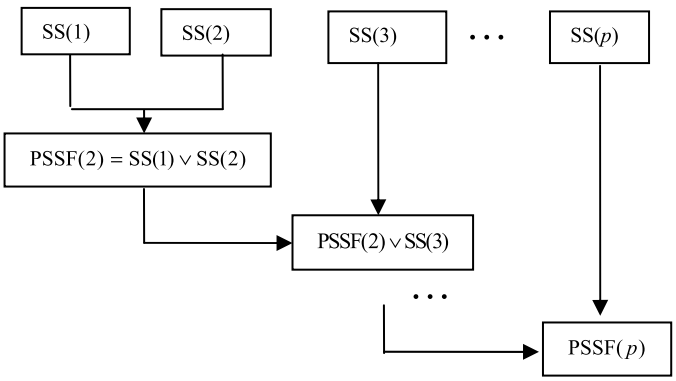


Fig. 4. Diagram of progressive fusion of p SC-SF classifiers for PMSSF method.

VI. Fusion of SVM-EPF and CEM-Gaussian Classifiers

This section presents an immediate application of Section II by fusing two particularly interesting SC-SF classifiers. One is the SVM-EPF classifiers in [11] which use an SVM classifier and an EPF as their respective SC and SF. The other is CEM-Gaussian classifiers which use CEM as a mixed pixel spectral classifier and a Gaussian filter as their SFs.

A. SVM-EPF Classifiers

Over the past years, SVM has been a major technique to be used as an SC. The drawback of using SVM is that SVM is a pure pixel and hard decision-made classifier. So, when it is used to perform classification, the spectral information of each data sample cannot be preserved. SVM performs spectral classification and then follows up an EPF to capture spatial contextual information from the SVM-obtained spectral classification map to produce probability maps for each of classes of interest that are used for final classification by MAP. There are four EPF-based techniques, EPF-B-c, EPF-G-c, EPF-B-g, and EPF-G-g, which were shown to be best classification techniques, where "B" and "G" are used to specify bilateral filter and guided filter respectively and "g" and "c" indicate that the first principal component and color composite of three principal components are used as reference images [11].

B. CEM-Gaussian Classifiers

Unlike many existing SC-SF classifiers methods reported in the literature including SVM-EPF, the CEM-Gaussian classifiers presented in this section take a different route to perform HSIC. Two key features distinguish CEM-Gaussian methods from other SC-SF classifiers. While most SC-SF classifiers use SVM as an SC, CEM-Gaussian method make use of subpixel detector CEM to perform spectral classification in conjunction with a Gaussian SF to extract spatial information from the CEM-classified maps for final classification.

C. Fusion of SVM-EPF and CEM-Gaussian Classifiers

For an illustrative purpose, this section presents an algorithm of fusing a pure pixel-based SVM-EPF classifier and a subpixel detection-based CEM-Gaussian classifier, referred to as iterative SVM-EPF $\vee$ CEM-Gaussian with detailed step-by-step implementation described as follows:

SVM-EPF $\vee$ CEM-Gaussian

1) Initial condition: Let $\Omega^{(0)} = \{\text{HSI}\}$. Set $k = 0$
2) At the kth iteration, implement EPF on $\Omega^{(k)}$ to produce a soft probability map of each class, C_i, $1 \le i \le M$ $\left\{\text{EPF}_i^{(k)}\right\}_{i=1}^{M}$.
3) Implement CEM on $\Omega^{(\text{k})}$ to generate M abundance fractional maps $\left\{\text{CEM}_i^{(k)}\right\}_{i=1}^{M}$.
4) Apply the same Gaussian filter to each of $\left\{\text{CEM}_i^{(k)}\right\}_{i=1}^{M}$ to produce Gaussian filtered $\left\{\text{GFCEM}_i^{(k)}\right\}_{i=1}^{M}$.
5) For each pixel (x, y) find the fused SFMap given by

$$\text{fused SFMap}_i^{(k)}(x,y) = \max\left\{\text{EPF}_i^{(k)}(x,y), \text{GFCEM}_i^{(k)}(x,y)\right\} \quad (8)$$

6) Use Otsu's method to threshold each of $\left\{\text{fused SFMap}_i^{(k)}\right\}_{i=1}^{M}$ to produce M binary images for classification maps, one for each class, $\left\{\text{BMAX}_i^{(k)}\right\}_{i=1}^{M}$.
7) If $k = 0$ or 1, go to step 10. Otherwise, continue.
8) Check TI for two cases
if

$$\text{TI}_i^{(k)} > \varepsilon \quad (9)$$

where

$$\text{TI}_i^{(k)} = \frac{|\text{BMAX}_i^{(k)} \cap \text{BMAX}_i^{(k-1)}|}{|\text{BMAX}_i^{(k)} \cup \text{BMAX}_i^{(k-1)}|} \text{ for } 1 \le i \le M \quad (10)$$

with $|S|$ being the size of set S, and go step 10,

$$\text{else } \text{TI}_i^{(k)} < \text{TI}_i^{(k-1)} \quad (11)$$

9) If yes, go to step 11. Otherwise, continue.
10) Let $k \leftarrow k + 1$ and form a new data cube

$$\Omega^{(k)} = \Omega^{(k-1)} \cup \left\{\text{fused SFMap}_1^{(k)}\right\} \cup \cdots \cup \left\{\text{fused SFMap}_M^{(k)}\right\} \quad (12)$$

Go to step 2.
11) SVM-EPF$\vee$CEM-Gaussian is terminated and $\{\text{BMAX}_{i=1}^{M}\}$ are the final classification maps.

Fig. 5 depicts a graphic diagram to implement SVM-EPF $\vee$ CEM-Gaussian.

VII. Classification Measures

It has been a common practice in HSI classification that OA, AA, and kappa coefficient are major measures to be used to evaluate classification performance. Unfortunately, it has been shown in [28] and [29] that OA and AA were not sufficiently enough to evaluate the effectiveness of a classifier. There is a

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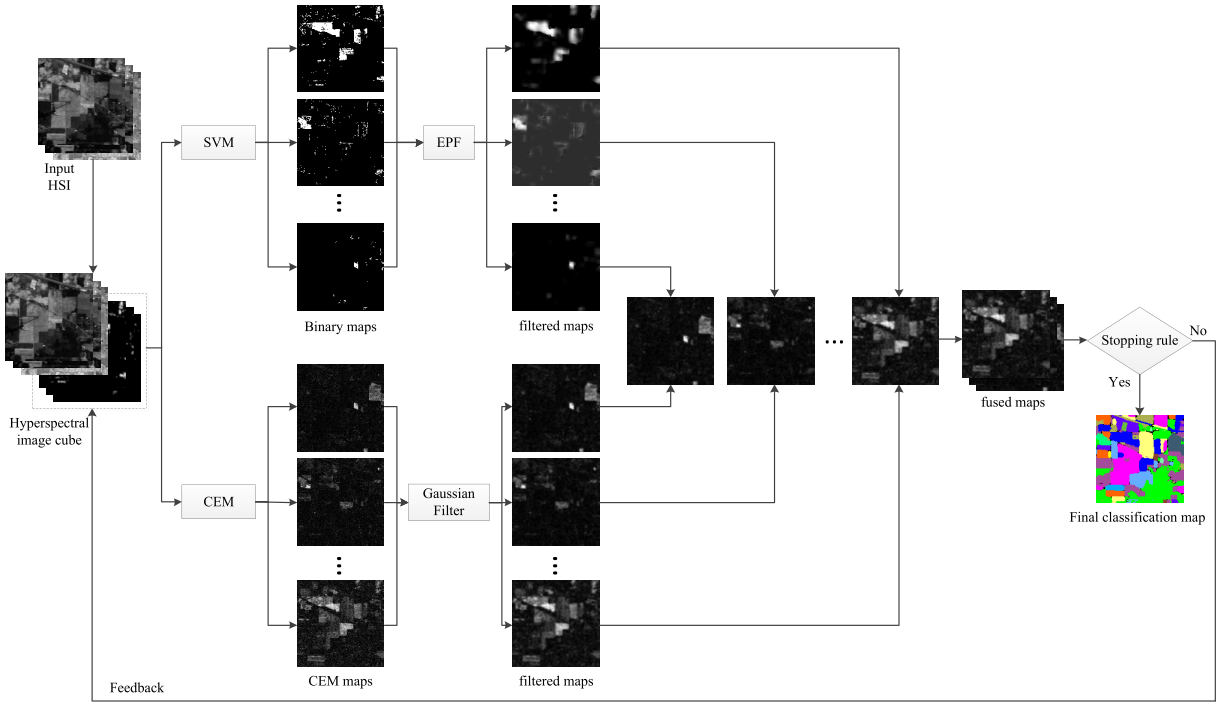


Fig. 5. Graphic diagram of iterative SVM-EPF ∨ CEM-Gaussian.

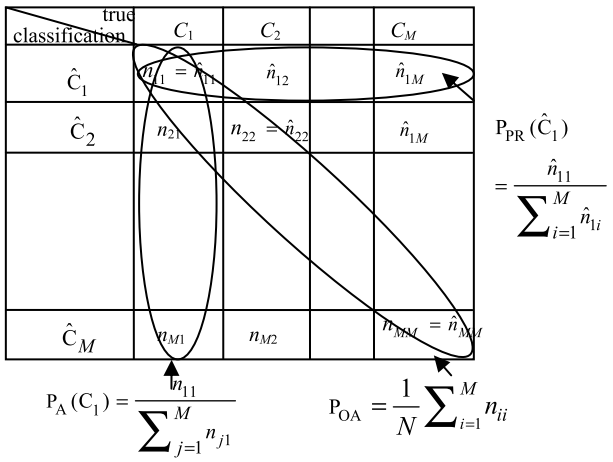


Fig. 6. M-class confusion matrix.

more important measure, called precision rate (PR), arising in pattern classification that can be used for this purpose.

In what follows, we provide precision definitions of each measure to be used for experiments [47]. Consider an M-class confusion matrix constructed in Fig. 6.

Several classification measures can be derived as follows:

1) M = the number of classes;
2) n_{ji} = the number of samples in the ith class to be classified into the jth class;
3) n_i = the number of data samples in the ith class, i.e., $n_i = \sum_{j=1}^{M} n_{ji}$;
4) $\hat{n}_{ji}$ = the number of data samples classified in the jth class, which are supposed to in the ith class, C_j;
5) $\hat{n}_j$ = the number of data samples classified in the ith class, i.e., $\hat{n}_j = \sum_{i=1}^{M} \hat{n}_{ji}$;
6) $\tilde{n}_i = \sum_{j=1, j\neq i}^{M} n_{ji}$ = the number of samples in the ith class with $i \neq j$ falsely classified into the jth class;
7) N = total number of data samples, $N = \sum_{j=1}^{M} n_j$;
8) C_i = set of samples in the ith class;
9) $\tilde{C}_i$ = set of data samples from classes $\{C_j\}_{j=1, j\neq i}^{M}$ classified into C_i;
10) $p(C_i) = (n_i/N)$.

In the following, we define several classification measures.

1) Accuracy

$$\begin{aligned} \mathrm{P_A}(C_i) &= \text{accuracy of the } j\text{th class} \\ &= \mathrm{P}(C_{ii}|C_i) = \frac{n_{ii}}{n_i}. \end{aligned} \tag{13}$$

2) OA

$$\begin{aligned} \mathrm{P_{OA}} &= p_{\text{overall-accuracy}}\left(\{C_i\}_{i=1}^{M}\right) \\ &= \sum_{i=1}^{M} p(\mathrm{C}_i)\, p(\mathrm{C}_{ii}|\mathrm{C}_i) = \sum_{j=1}^{M} \left(\frac{n_i}{N}\right)\left(\frac{n_{ii}}{\sum_{j=1}^{M} n_{ji}}\right) \\ &= \frac{1}{N}\sum_{i=1}^{M} n_{ii}. \end{aligned} \tag{14}$$

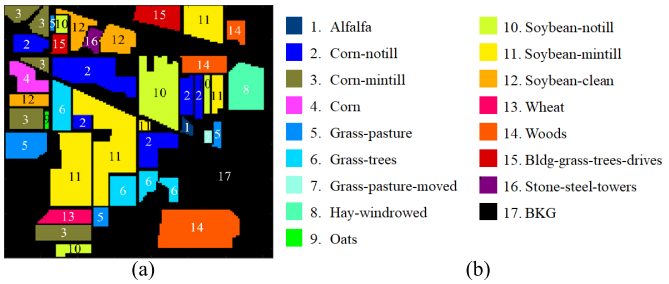
3) AA

$$\mathrm{P_{AA}} = \frac{1}{M}\sum_{i=1}^{M} \mathrm{P_A}(C_i) = \frac{1}{M}\sum_{i=1}^{M} \frac{n_{ii}}{n_i}. \tag{15}$$

4) Misclassification rate

$$\begin{aligned} p(\tilde{C}_i|C_i) &= \text{misclassification rate of the } i\text{th class}, C_i \\ &= \sum_{j=1}^{M} p(C_{ji}|C_i) = \frac{\sum_{j=1, j\neq i}^{M} n_{ji}}{\sum_{j=1}^{M} n_{ji}} \\ &= 1 - p(C_{ii}|C_i) = 1 - \mathrm{P_A}(C_i) \end{aligned} \tag{16}$$

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(a) (b)

Fig. 7. Purdue's Indiana Indian Pines scene. (a) Ground-truth map. (b) Classes by colors.

$$\mathrm{P}(\tilde{C}) = \sum_{i=1}^{M} p(C_i)p(\tilde{C}_i|C_i) = \text{misclassification rate}. \tag{17}$$

5) PR

$$\mathrm{P}_{\mathrm{PR}}(\hat{C}_i) = p(\hat{C}_i|\text{classification}) = \frac{\hat{n}_{ii}}{\sum_{j=1}^{M}\hat{n}_{ji}} \tag{18}$$

$$\mathrm{P}_{\mathrm{APR}} = \frac{1}{M}\sum_{i=1}^{M}\mathrm{P}_{\mathrm{PR}}(\hat{C}_i). \tag{19}$$

It should be noted that P_{PR} in (17) is generally calculated for the entire image scene including BKG. If P_{PR} in (17) is calculated only for classes of interest without accounting for BKG, then for each class, C_i, n_{ii} in the diagonal line in the confusion matrix in Fig. 6 is identical to $\hat{n}_{ii}$and $\hat{N} = \sum_{i=1}^{M}\hat{n}_i$ equal to $N = \sum_{i=1}^{M} n_i$. So

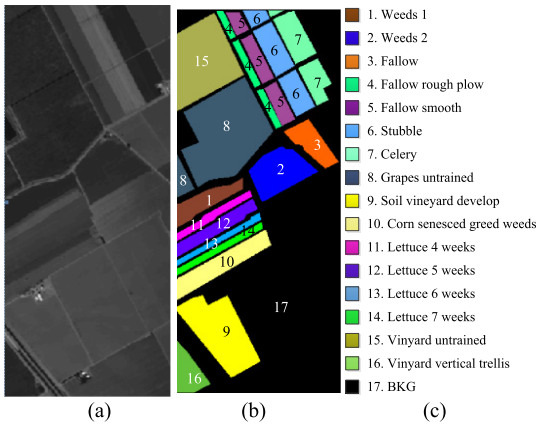
$$\mathrm{P}_{\mathrm{OPR}} = \sum_{i=1}^{M} p(\hat{C}_i)\mathrm{P}_{\mathrm{PR}}(\hat{C}_i) = \sum_{i=1}^{M}\left(\frac{\hat{n}_i}{\hat{N}}\right)\mathrm{P}_{\mathrm{PR}}(\hat{C}_i)$$
$$= \sum_{i=1}^{M}\left(\frac{n_i}{N}\right)\mathrm{P}_{\mathrm{A}}(C_i) = \mathrm{P}_{\mathrm{OA}} \tag{20}$$

VIII. Real Images to Be Studied

Four widely used real hyperspectral images, Purdue University's Indiana Indian Pines, Salinas, University of Pavia and Houston, available at the website http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes and https://hyperspectral.ee.uh.edu/?page_id=459 were used for experiments to demonstrate superiority of fused SC-SF classifier in classification compared to the currently being used SC-SF classifiers.

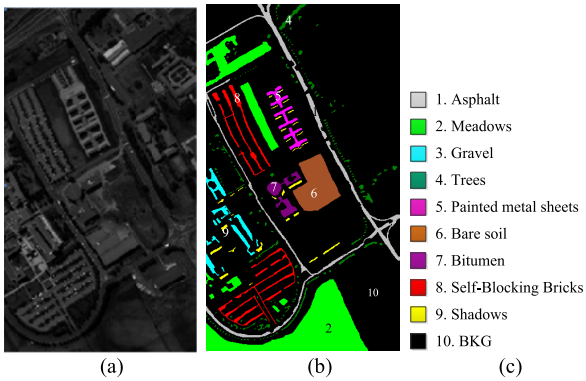
A. Purdue Indiana Indian Pines Data

The first data set used for experiments is a well-known AVIRIS image scene, Purdue Indiana Indian Pines test site shown in Fig. 7(a) with its ground truth of 17 class maps including BKG in Fig. 7(b). It has size of 145 × 145 pixel vectors taken from an area of mixed agriculture and forestry in Northwestern Indiana, USA with details of band and wavelength is given in caption. The data set is available at website http://cobweb.ecn.purdue.edu/~biehl/MultiSpec/documentation.html. It was recorded in June 1992 with 220 bands which include water absorption bands (bands 104–108 and 150–163, 220).



(a) (b) (c)

Fig. 8. Ground-truth of Salinas scene with 16 classes. (a) Salinas scene. (b) Ground-truth image. (c) Classes by colors.



(a) (b) (c)

Fig. 9. Ground-truth of University of Pavia scene with nine classes. (a) University of Pavia scene. (b) Ground-truth map. (c) Classes by colors.

B. Salinas Data

A second AVIRIS data used for experiments was Salinas scene shown in Fig. 8(a), which was captured by the AVIRIS sensor over Salinas Valley, CA, USA, with a spatial resolution of 3.7 m per pixel and a spectral resolution of 10 nm. It has size of 512 × 217 × 224. Fig. 8(b) and (c) shows the color composite of the Salinas image along with the corresponding ground truth class labels.

C. ROSIS Data

A third hyperspectral image data used for experiments was the University of Pavia image shown in Fig. 9 which is an urban area surrounding the University of Pavia, Italy. It was recorded by the ROSIS-03 satellite sensor. It is of size 610 × 340 × 115 with a spatial resolution of 1.3 m per pixel and a spectral coverage ranging from 0.43 to 0.86 μm with spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Nine classes of interest are considered for this image.

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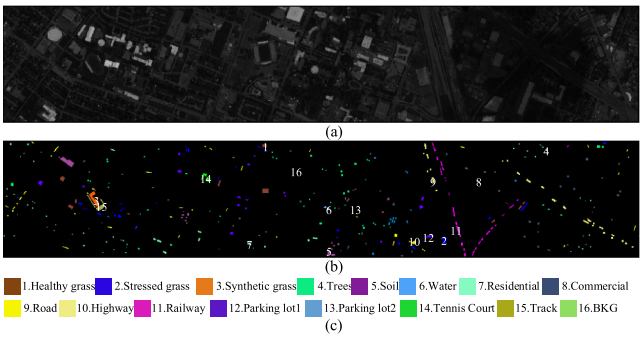


Fig. 10. Houston scene along with ground-truth map of 15 classes. (a) Houston scene. (b) Ground-truth time. (c) Classes by colors.

D. Houston Data

A fourth image scene was collected over the campus area of University of Houston and its neighboring urban area as shown in Fig. 10. The image was captured by compact airborne spectrographic image (CASI) sensor in 2012. The data set was published in 2013 IEEE GRSS Data Fusion Contest. It has size of 349 × 1905 × 144 with 144 spectral bands ranging 380–1050 nm and spatial resolution of 2.5 m.

IX. Experimental Results and Discussion

To validate the utility of fusing SC-SF classifiers, the following three different types of classifiers were used and evaluated for a comparative study and analysis.

1) *Pixel-Based Classifiers:* SVM-EPF-G-g [11] and ISVM-EPF-G-g which is an iterative version of EPF-G-g developed in [48], referred to as iterative EPF-G-g (IEPF-G-g).
2) *Mixed-Pixel Classifiers:* CEM-Gaussian and ICEM-Gaussian developed in [28] which is an iterative version of CEM coupled with Gaussian filters.
3) *Feature-Based Classifiers:* OTVCA [49], [50], which is a state-of-the-art SS reconstruction-based feature extraction method to extract features by minimizing a total variation penalized optimization problem. In addition, a new SC-SF extended version of OTVCA, called OTVCA-Gaussian, is further developed in this section which implements OTVCA in conjunction with Gaussian filters to improve OTVCA. This OTVCA-Gaussian is particularly derived from a similar idea proposed in [27].

Table I tabulates the number of training samples, n_{train} and number of test samples, n_{test} used for experiments conducted for Purdue's data, Salinas University of Pavia and Houston data in this section.

TABLE I

Numbers of Training Samples, n_{train} and Numbers of Test Samples, n_{test} Used for Purdue's Data, Salinas Data, University of Pavia Data and Houston Data

	Purdue		Salinas		University of Pavia		Houston	
Class	n_{train}	n_{test}	n_{train}	n_{test}	n_{train}	n_{test}	n_{train}	n_{test}
1	25	21	50	1909	50	6531	50	1003
2	83	1345	50	3626	50	18549	50	1014
3	78	752	50	1876	50	1999	50	455
4	68	169	50	1292	50	2964	50	1006
5	79	404	50	2578	50	1245	50	1006
6	78	652	50	3859	50	4929	50	93
7	14	14	50	3479	50	1230	50	1022
8	66	412	50	11171	50	3582	50	1003
9	10	10	50	6103	50	847	50	1009
10	81	891	50	3178			50	986
11	99	2356	50	968			50	1004
12	73	520	50	1827			50	991
13	70	135	50	816		-	50	235
14	90	1175	50	970			50	197
15	65	321	50	7168			50	423
16	46	47	50	1707				-
Total	1025	9224	800	52527	450	41876	750	11447

A. Fusion of Two SS Classifiers

Tables II–V tabulate P_A, P_{PR}, and P_{PR}(without BKG) produced by single classifiers including OTVCA, OTVCA-Gaussian, IEPF, and ICEM-Gaussian for the four image scenes where the best results were highlighted and boldfaced by black. In particular, $P_{APR} = P_{AA}$ were calculated by (19) and $P_{OPR} = P_{OA}$ calculated by (20) in two versions. One is to account for BKG pixels which were misclassified into various classes. The other is to exclude all BKG pixels from P_{PR} calculation, denoted by P_{PR} (without BKG), which was calculated only for classes of interest without including BKG classes, that is, 16 classes for both the Purdue and Salinas data sets, nine classes for University of Pavia, and 15 classes for Houston data. In this case, the best results for P_{PR} (without BKG) were boldfaced and highlighted by red. The running time was further calculated in seconds and documented in at the second last row of the tables with the computer environment specified by an Intel Core i5 with 2.9 GHz and 8-GB RAM. Finally, the numbers of iterations, n_{iter}, were also listed at the last row of tables.

Tables VI–IX also tabulate P_A, P_{PR}, and P_{PR}(without BKG) produced by fusing two different classifiers, i.e., OTVCA $\vee$ SVM-EPF-G-g, OTVCA $\vee$ CEM-Gaussian and SVM-EPF-G-g $\vee$ CEM-Gaussian, respectively. As shown in Tables VI–IX, the best results of P_{AA} and P_{OA} were produced by SVM-EPF-G-g $\vee$ CEM-Gaussian where OTVCA was improved by its SC-SF version, OTVCA-Gaussian, by 1.5%, 0.1%, 0.8%, and 2.2% in terms of P_{OA} in the four data sets. Interestingly, such SC-SF OTVCA version, OTVCA-Gaussian, could further be improved by OTVCA $\vee$ SVM-EPF-G-g and OTVCA $\vee$ CEM-Gaussian in all the four data sets. Nevertheless, OTVCA-Gaussian fused with SVM-G-g and CEM-Gaussian still could not compete against SVM-G-g and CEM-Gaussian in all the four data sets. This is mainly due to the fact that OTVCA is a feature-based classifier compared to SVM-G-g which is a pixel-based classifier. This is exactly the same as fusion decision where feature-level fusion is generally is not good as data sample-level fusion because the former loses sample information which is included into extracted features.

As for P_{PR} accounting for BKG, ICEM-Gaussian was always the best to produce highest value of P_{PR} (with BKG) better than that produced by OTVCA, OTVCA-Gaussian, ISVM-EPF-G-g and fusing any two classifiers for all the four

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TABLE II

P_A, P_{PR} AND P_{PR}(WITHOUT BKG) PRODUCED BY SINGLE CLASSIFIERS INCLUDING OTVCA, OTVCA-GAUSSIAN, ISVM-EPF-G-G AND ICEM-GAUSSIAN FOR PURDUE DATA WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA			**OTVCA-Gaussian**			**ISVM-EPF-G-g**			**ICEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	1.0000	0.5169	1.0000	0.9304	0.6360	0.9529	0.9783	0.3982	1.0000	0.9783	0.6000	1.0000
2	0.9559	0.6500	0.9743	0.9679	0.5752	0.9691	0.9629	0.5934	0.9528	0.9608	0.8695	1.0000
3	0.9566	0.7620	0.9601	0.9855	0.6033	0.9845	0.9880	0.5636	0.9832	0.9614	0.8347	0.9913
4	1.0000	0.6810	0.9556	1.0000	0.4216	0.9575	1.0000	0.7226	0.9834	0.9873	0.8327	1.0000
5	0.9814	0.1922	0.9979	0.9607	0.5147	0.9547	0.9896	0.1905	0.9735	0.9275	0.8485	1.0000
6	1.0000	0.6202	1.0000	0.9932	0.6000	0.9667	1.0000	0.5620	1.0000	0.9699	0.8281	1.0000
7	0.9643	0.5400	1.0000	1.0000	0.1647	0.9655	1.0000	0.1069	0.9655	1.0000	0.5091	1.0000
8	1.0000	0.7399	1.0000	0.9812	0.8031	0.9958	1.0000	0.7231	1.0000	0.9895	0.9167	1.0000
9	1.0000	0.2985	0.9524	0.9500	0.5043	0.9548	1.0000	0.2500	0.9524	1.0000	0.3571	0.9091
10	0.9259	0.7103	0.8824	0.9572	0.5530	0.9872	0.9815	0.7327	0.9199	0.9259	0.8563	0.9934
11	0.9242	0.7961	0.9598	0.9668	0.6311	0.9597	0.9356	0.8629	0.9837	0.9723	0.9353	0.9983
12	0.9680	0.6457	0.9456	0.9870	0.4315	0.9440	0.9966	0.6640	0.9752	0.9427	0.8653	0.9982
13	0.9951	0.7612	1.0000	1.0000	0.5511	0.9762	1.0000	0.6788	0.9951	0.9854	0.8279	1.0000
14	0.9368	0.2856	0.9992	0.9810	0.3234	0.9313	1.0000	0.3021	0.9968	0.9249	0.9070	1.0000
15	1.0000	0.1775	0.8301	0.9813	0.2848	0.9767	0.9896	0.1510	0.9646	0.9534	0.8846	1.0000
16	0.9892	0.7244	1.0000	0.9699	0.6742	1.0000	0.9785	0.7054	1.0000	0.9892	0.7541	0.9787
P_{AA}|P_{APR}	0.9748	0.5688	0.9661	0.9758	0.5170	0.9673	**0.9875**	0.5130	0.9779	0.9668	**0.7892**	**0.9918**
P_{OA}=P_{OPR}	0.9443			0.9590			**0.9753**			0.9566		
t(s)	33.6			34.1			50.8			44.0		
n_{iter}	-			-			9			8		

TABLE III

P_A, P_{PR} AND P_{PR}(WITHOUT BKG) PRODUCED BY OTVCA, OTVCA-GAUSSIAN, ISVM-EPF-G-G AND ICEM-GAUSSIAN FOR SALINAS WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA			**OTVCA-Gaussian**			**ISVM-EPF-G-g**			**ICEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	0.9965	0.8021	0.9990	0.9866	0.8011	0.9890	1.0000	0.7680	1.0000	0.9771	0.9392	1.0000
2	0.9987	0.5225	0.9949	0.9874	0.5336	0.9919	0.9949	0.5508	1.0000	0.9911	0.9775	1.0000
3	0.9980	0.0947	0.9513	0.9954	0.0926	0.9353	1.0000	0.0858	1.0000	0.9474	0.9185	1.0000
4	0.9950	0.1867	0.9921	0.9957	0.1819	0.9754	0.9978	0.1711	0.9907	0.9677	0.8226	0.9963
5	0.9754	0.5574	0.9950	0.9690	0.5594	0.9969	0.9937	0.5129	0.9970	0.9698	0.8924	1.0000
6	0.9962	0.7696	0.9972	0.9995	0.6851	1.0000	0.9995	0.8211	1.0000	0.9879	0.9179	1.0000
7	0.9969	0.8528	0.9992	0.9969	0.8485	0.9961	0.9939	0.8445	1.0000	0.9897	0.8978	1.0000
8	0.8366	0.7918	0.8199	0.8546	0.7549	0.8126	0.8930	0.8089	0.9482	0.9823	0.9815	1.0000
9	0.9911	0.3140	0.9935	0.9918	0.3831	0.9903	1.0000	0.4281	0.9992	0.9615	0.9909	1.0000
10	0.9384	0.4732	0.8653	0.9384	0.5514	0.8849	0.9851	0.3622	0.9917	0.9790	0.9742	1.0000
11	0.9850	0.2417	0.9343	0.9682	0.1666	0.8815	0.9972	0.2802	1.0000	0.9906	0.8560	1.0000
12	0.9979	0.6254	0.9683	1.0000	0.3811	0.9807	1.0000	0.7805	1.0000	0.9818	0.9436	1.0000
13	0.9869	0.6551	0.9891	0.9771	0.6285	0.9697	0.9956	0.5355	0.9785	0.9771	0.8672	1.0000
14	0.9879	0.3344	0.8329	0.9589	0.5516	0.9491	0.9907	0.5316	0.9364	0.9888	0.8780	1.0000
15	0.6924	0.7279	0.7833	0.6875	0.7694	0.7989	0.9242	0.8158	0.8491	0.9824	0.9917	1.0000
16	0.9867	0.7545	0.9900	0.9911	0.6392	0.9900	1.0000	0.7732	0.9994	0.9834	0.9549	1.0000
P_{AA}|P_{APR}	0.9600	0.5440	0.9441	0.9561	0.5530	0.9464	**0.9853**	0.5669	0.9806	0.9786	**0.9252**	**0.9998**
P_{OA}=P_{OPR}	0.9165			0.9175			0.9652			**0.9790**		
t(s)	97.5			98.7			71.1			319.1		
n_{iter}	-			-			12			16		

data sets. In particular, for University of Pavia CEM could take care of BKG by inverting the sample correlation matrix $\mathbf{R}$, i.e., $\mathbf{R}^{-1}$, to suppress BKG. When it was fused with OTVCA, OTVCA-Gaussian, SVM-G-g, its fusion performance was degraded compared to ICEM-Gaussian implemented alone but could be still improved over OTVCA, OTVCA-Gaussian,

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TABLE IV

P_A, P_{PR} AND P_{PR}(WITHOUT BKG) PRODUCED BY SINGLE CLASSIFIERS INCLUDING OTVCA, OTVCA-GAUSSIAN, ISVM-EPF-G-G AND ICEM-GAUSSIAN FOR UNIVERSITY OF PAVIA WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA			**OTVCA-Gaussian**			**ISVM-EPF-G-g**			**ICEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	0.8769	0.1541	0.9522	0.8227	0.2456	0.9906	0.9489	0.2070	0.9976	0.8459	0.3159	0.9919
2	0.9329	0.3256	0.9588	0.8533	0.3500	0.9640	0.9789	0.3783	0.9940	0.8338	0.8243	0.9993
3	0.8132	0.3144	0.7682	0.8913	0.2610	0.9410	0.9671	0.1965	0.9585	0.8552	0.4479	0.9803
4	0.9373	0.1019	0.8802	0.9455	0.1022	0.8867	0.9889	0.0728	0.9398	0.7800	0.1844	0.9469
5	0.9970	0.2569	0.9457	1.0000	0.3580	0.9985	0.9993	0.2830	0.9919	0.9985	0.6044	1.0000
6	0.8640	0.1003	0.8270	0.9248	0.0993	0.5802	0.9825	0.1398	0.9589	0.9268	0.8962	1.0000
7	0.9323	0.2341	0.6927	0.9917	0.3598	0.9459	1.0000	0.3305	1.0000	0.9699	0.6862	1.0000
8	0.8373	0.1808	0.8493	0.9669	0.1754	0.8975	0.9905	0.1799	0.9177	0.8772	0.3300	0.9338
9	0.9989	0.0806	0.9989	0.9979	0.0758	0.9968	1.0000	0.0765	0.9979	0.8944	0.2235	0.9988
$P_{AA}\|P_{APR}$	0.9100	0.1943	0.8748	0.9327	0.2252	0.9112	**0.9840**	0.2071	0.9729	0.8869	**0.5014**	**0.9834**
P_{OA}=P_{OPR}	0.9058			0.9139			**0.9776**			0.8583		
t(s)	102.9			103.5			75.8			170.0		
n_{iter}	-			-			12			13		

TABLE V

P_A, P_{PR}, AND P_{PR}(WITHOUT BKG) PRODUCED BY SINGLE CLASSIFIERS INCLUDING OTVCA, OTVCA-GAUSSIAN, ISVM-EPF-G-G AND ICEM-GAUSSIAN FOR HOUSTON DATA WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA			**OTVCA-Gaussian**			**ISVM-EPF-G-g**			**ICEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	0.9459	0.0128	0.9113	0.8810	0.0185	0.9552	0.9164	0.0277	0.9757	0.7692	0.1158	0.9689
2	0.9699	0.0091	0.9529	0.9915	0.0072	0.8960	0.9812	0.0165	0.9500	0.7378	0.0747	0.9937
3	0.9842	0.3276	1.0000	1.0000	0.4970	1.0000	1.0000	0.4870	1.0000	0.9941	0.5553	1.0000
4	0.9328	0.0164	0.9830	0.9290	0.0140	0.9761	0.9905	0.0094	0.9518	0.8797	0.0458	0.9718
5	0.9716	0.0211	0.9088	1.0000	0.0248	0.9609	1.0000	0.0542	0.9832	0.9280	0.1799	0.9589
6	0.9580	0.0924	0.8354	0.9860	0.0069	0.7972	0.9231	0.1587	1.0000	0.7972	0.2192	1.0000
7	0.8498	0.0145	0.7826	0.8088	0.0133	0.7404	0.9888	0.0111	0.9541	0.7845	0.0264	0.8928
8	0.7217	0.0141	0.9069	0.5242	0.0148	0.8351	0.8651	0.0134	0.9902	0.6201	0.0315	0.9939
9	0.7885	0.0112	0.7489	0.7828	0.0088	0.8210	0.9802	0.0079	0.9629	0.8121	0.0237	0.9399
10	0.8851	0.0195	0.7745	0.9305	0.0185	0.7633	1.0000	0.0486	1.0000	0.9720	0.0289	0.8498
11	0.6357	0.0336	0.8590	0.7592	0.0232	0.7847	1.0000	0.0274	0.9981	0.8273	0.0477	0.9625
12	0.8223	0.0186	0.7963	0.8352	0.0216	0.7588	0.9635	0.0340	0.9766	0.8261	0.0356	0.8776
13	0.5684	0.0051	0.5891	0.7404	0.0066	0.7662	0.9649	0.0069	0.8359	0.8912	0.0274	0.9338
14	0.9960	0.0102	0.7432	1.0000	0.0225	0.8792	1.0000	0.1942	0.9251	0.9676	0.4509	1.0000
15	0.9852	0.2733	0.9979	1.0000	0.2055	1.0000	1.0000	0.2470	1.0000	1.0000	0.3551	0.9958
$P_{AA}\|P_{APR}$	0.8677	0.0586	0.8526	0.8779	0.0602	0.8623	**0.9716**	0.0896	**0.9669**	0.8538	**0.1479**	0.9559
P_{OA}=P_{OPR}	0.8605			0.8825			**0.9711**			0.8345		
t(s)	695.7			698.4			193.7			569.4		
n_{iter}	-			-			12			10		

SVM-G-g, which were implemented alone. For example, P_{PR} (with BKG) of ISVM-EPF-G-g could be only improved very little by 2%–5%. This indicated that BKG was still a serious issue for EPF and fusing ICEM could do very little on improvement. Most interestingly, comparing P_{PR} (with BKG) to P_{PR} (without BKG), P_{PR} (with BKG) was only half of P_{PR} (without BKG) for the Purdue and Salinas data sets but much worse for University of Pavia and Houston with P_{PR} (with BKG) being only 1/4 and 1/10 of P_{PR} (without BKG) for ISVM-EPF-G-g. These results provided evidence that the BKG is indeed an important and serious issue that cannot be neglected or overlooked and must be addressed in classification.

Figs. 11–15 show classification maps of all the test methods for the four data sets. By visual inspection, the worst results are apparently produced by OTVCA. Also, the fused classification maps obtained by various fusion methods indeed improved results produced by using single SC-SF classifiers alone.

B. Simultaneous Fusion of Three SS Classifiers

In this section, three SC-SF classifiers, SVM-EPF-G-g, CEM-Gaussian, and OTVCA-Gaussian, were fused

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TABLE VI

P_A, P_{PR} AND P_{PR} (WITHOUT BKG) PRODUCED BY OTVCA ∨ SVM-EPF-G-G, OTVCA ∨ CEM-GAUSSIAN AND SVM-EPF-G-G ∨ CEM-GAUSSIAN FOR PURDUE DATA WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA-Gaussian ∨ SVM-EPF-G-g			**OTVCA-Gaussian ∨ CEM-Gaussian**			**SVM-EPF-G-g ∨ CEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	1.0000	0.3459	1.0000	0.9783	0.3719	1.0000	1.0000	0.3833	1.0000
2	0.9398	0.7188	0.9690	0.9573	0.6225	0.9856	0.9713	0.6536	0.9971
3	0.9988	0.5307	0.9776	0.9904	0.6362	0.9693	0.9855	0.7560	1.0000
4	1.0000	0.6771	1.0000	1.0000	0.7182	1.0000	1.0000	0.7383	0.9916
5	0.9959	0.2301	0.9757	0.9959	0.2395	0.9959	0.9959	0.1502	0.9897
6	1.0000	0.5753	1.0000	0.9959	0.5601	0.9986	1.0000	0.5534	1.0000
7	1.0000	0.1443	0.9655	1.0000	0.1573	1.0000	0.9643	0.3699	1.0000
8	1.0000	0.7231	1.0000	1.0000	0.7199	1.0000	1.0000	0.7040	1.0000
9	1.0000	0.1015	0.9524	1.0000	0.0664	0.8333	1.0000	0.3846	1.0000
10	0.9599	0.6377	0.8920	0.9702	0.7221	0.9300	0.9928	0.6108	0.9297
11	0.9483	0.8229	0.9798	0.9633	0.8028	0.9821	0.9792	0.8166	0.9963
12	0.9983	0.7023	0.9900	0.9949	0.6592	0.9850	0.9966	0.6762	0.9883
13	1.0000	0.7455	1.0000	1.0000	0.6926	1.0000	1.0000	0.7168	1.0000
14	1.0000	0.3061	0.9976	0.9779	0.2850	0.9960	1.0000	0.2604	0.9992
15	1.0000	0.1320	0.9897	0.9922	0.1631	0.9387	0.9974	0.3941	0.9974
16	0.9892	0.6715	1.0000	1.0000	0.6643	0.9894	0.9892	0.7797	0.9892
P_{AA}|P_{APR}	0.9894	0.5041	0.9806	0.9885	0.5051	0.9752	**0.9920**	**0.5592**	**0.9924**
P_{OA}=P_{OPR}	0.9649			0.9678			**0.9885**		
t(s)	359.8			384.7			84.3		
n_{iter}	11			12			11		

TABLE VII

P_A, P_{PR} AND P_{PR} (WITHOUT BKG) PRODUCED BY OTVCA ∨ SVM-EPF-G-G, OTVCA ∨ CEM-GAUSSIAN AND SVM-EPF-G-G ∨ CEM-GAUSSIAN FOR SALINAS WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA-Gaussian ∨ SVM-EPF-G-g			**OTVCA-Gaussian ∨ CEM-Gaussian**			**SVM-EPF-G-g ∨ CEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	0.9950	0.7706	1.0000	0.9960	0.7832	1.0000	1.0000	0.7650	1.0000
2	0.9968	0.5502	0.9973	0.9970	0.5515	0.9979	0.9981	0.5510	1.0000
3	0.9960	0.0825	0.9870	0.9995	0.0846	1.0000	1.0000	0.0957	1.0000
4	0.9928	0.1877	0.9921	0.9964	0.1860	0.9686	0.9928	0.1846	0.9830
5	0.9944	0.5507	0.9959	0.9836	0.5812	0.9895	0.9922	0.5811	0.9944
6	0.9997	0.7464	1.0000	0.9997	0.7387	1.0000	0.9997	0.8215	1.0000
7	0.9953	0.8457	1.0000	0.9980	0.8371	1.0000	0.9975	0.8310	1.0000
8	0.8678	0.8907	0.9529	0.8879	0.9107	0.9656	0.9508	0.8829	0.9983
9	0.9997	0.4322	0.9998	0.9982	0.4566	0.9997	0.9995	0.4878	0.9998
10	0.9887	0.3445	0.9457	0.9973	0.3463	0.9855	0.9966	0.2440	0.9912
11	0.9963	0.2924	1.0000	0.9963	0.2521	0.9991	0.9963	0.2690	1.0000
12	1.0000	0.7620	1.0000	1.0000	0.7341	0.9995	1.0000	0.7748	1.0000
13	0.9978	0.5851	0.9870	0.9978	0.6432	1.0000	0.9934	0.6124	0.9956
14	0.9907	0.6261	0.9779	0.9916	0.5921	0.9707	0.9850	0.6518	0.9777
15	0.9313	0.7631	0.8350	0.9465	0.7642	0.8477	0.9957	0.8574	0.9302
16	1.0000	0.7061	0.9972	0.9939	0.7319	1.0000	1.0000	0.7292	1.0000
P_{AA}|P_{APR}	0.9839	0.5710	0.9792	0.9862	0.5746	0.9827	**0.9936**	**0.5837**	**0.9919**
P_{OA}=P_{OPR}	0.9609			0.9672			**0.9876**		
t(s)	527.3			569.2			206.3		
n_{iter}	6			6			11		

simultaneously. Table X tabulate P_A, P_{PR}, and P_{PR} (without BKG) produced by SVM-EPF ∨ CEM-Gaussian ∨ OTVCA-Gaussian using one-against-all and one-against-one for the four image scenes where the best results are boldfaced. In these tables, the running times in seconds and the number of iterations, n_{iter}, are provided at the last two rows of tables.

From Table X it is very clear to show that the best simultaneous fusion results were always produced by the one-against-one strategy. Comparing the results in Table X to the results in Tables II–IX, the P_{OA} of four SC-SF classifiers using the one-against-one method was further improved by 0.07%, 0.17%, 0.11%, and 0.42%. Although these results were not

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TABLE VIII

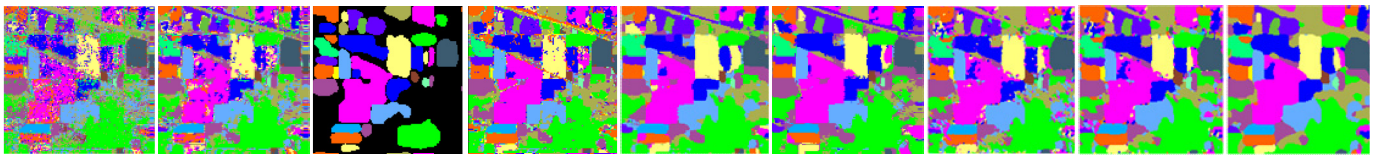
P_A, P_{PR}, AND P_{PR} (WITHOUT BKG) PRODUCED BY OTVCA ∨ SVM-EPF-G-G, OTVCA ∨ CEM-GAUSSIAN AND SVM-EPF-G-G ∨ CEM-GAUSSIAN FOR UNIVERSITY OF PAVIA WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA-Gaussian ∨ SVM-EPF-G-g			**OTVCA-Gaussian ∨ CEM-Gaussian**			**SVM-EPF-G-g ∨ CEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	0.9656	0.1686	0.8521	0.9647	0.1604	0.8454	0.9875	0.2054	0.9962
2	0.9769	0.4296	0.9945	0.9680	0.3814	0.9920	0.9795	0.4908	0.9990
3	0.6341	0.2078	0.9576	0.6355	0.2127	0.9481	0.9590	0.2714	1.0000
4	0.9119	0.1119	0.9503	0.9259	0.0862	0.8581	0.9899	0.0655	0.9141
5	0.9993	0.3933	0.9993	0.9993	0.4694	1.0000	1.0000	0.3511	0.9993
6	1.0000	0.1103	0.9482	0.9938	0.1244	0.9664	1.0000	0.1194	0.9859
7	0.4880	0.2271	0.9908	0.5098	0.2428	0.9912	1.0000	0.3682	1.0000
8	0.9112	0.1689	0.7772	0.9017	0.1928	0.7839	0.9954	0.1713	0.9527
9	0.9261	0.0915	0.9744	0.8712	0.1214	0.9988	0.9979	0.1165	1.0000
P_{AA}|P_{APR}	0.8681	0.2121	0.9383	0.8633	0.2213	0.9315	**0.9899**	**0.2400**	**0.9830**
P_{OA}=P_{OPR}	0.9351			0.9301			**0.9859**		
t(s)	1094.0			1065.7			323.3		
n_{iter}	12			11			18		

TABLE IX

P_A, P_{PR}, AND P_{PR} (WITHOUT BKG) PRODUCED BY OTVCA ∨ SVM-EPF-G-G, OTVCA ∨ CEM-GAUSSIAN AND SVM-EPF-G-G ∨ CEM-GAUSSIAN FOR HOUSTON DATA WITH RUNNING TIME t AND NUMBERS OF ITERATIONS

	OTVCA-Gaussian ∨ SVM-EPF-G-g			**OTVCA-Gaussian ∨ CEM-Gaussian**			**SVM-EPF-G-g ∨ CEM-Gaussian**		
	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)	P_A	P_{PR}	P_{PR}(w/o BKG)
1	0.9240	0.0138	0.9789	0.9231	0.0254	0.9524	0.9240	0.0267	0.9605
2	0.9615	0.0100	0.9859	0.9709	0.0129	0.9377	0.9615	0.0159	0.9481
3	1.0000	0.4208	0.9980	1.0000	0.4825	1.0000	1.0000	0.4889	1.0000
4	1.0000	0.0137	0.9757	1.0000	0.0489	0.9428	1.0000	0.0099	0.9531
5	0.9981	0.0234	0.9875	0.9981	0.0549	0.9715	0.9981	0.0552	0.9805
6	0.8462	0.1969	0.9859	0.8671	0.1816	1.0000	0.8462	0.1726	1.0000
7	0.9701	0.0184	0.9236	0.9692	0.0089	0.9767	0.9701	0.0125	0.9858
8	0.9307	0.0156	0.9267	0.9307	0.0036	0.9900	0.9307	0.0162	1.0000
9	0.9802	0.0112	0.8510	0.9802	0.0100	0.9476	0.9802	0.0101	0.9593
10	0.9826	0.0245	0.8502	0.9797	0.0296	1.0000	0.9826	0.0484	1.0000
11	0.9972	0.0270	0.9210	0.9962	0.0157	0.9609	0.9972	0.0154	0.9705
12	0.9933	0.0199	0.9740	0.9894	0.0221	0.9823	0.9933	0.0318	0.9914
13	0.9509	0.0053	0.9115	0.9474	0.0034	0.8482	0.9509	0.0064	0.8686
14	1.0000	0.0133	0.9609	1.0000	0.3536	1.0000	1.0000	0.3946	1.0000
15	1.0000	0.2810	1.0000	1.0000	0.3050	1.0000	1.0000	0.3079	1.0000
P_{AA}|P_{APR}	0.9690	0.0930	0.9636	0.9701	0.0925	0.9606	**0.9790**	**0.1075**	**0.9745**
P_{OA}=P_{OPR}	0.9703			0.9675			**0.9763**		
t(s)	7485.8			7297.5			726.4		
n_{iter}	10			9			13		



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 11. Classification maps for Purdue data. (a) OTVCA. (b) ISVM-EPF-G-g. (c) ICEM-Gaussian. (d) OTVCA ∨ SVM-EPF-G-g. (e) OTVCA ∨ SVM-EPF-G-g. (f) SVM-EPF-G-g ∨ CEM-Gaussian. (g) and (h) SVM-EPF ∨ CEM-Gaussian ∨ OTVCA-Gaussian using one-against-all and one-against-one. (i) Progressive fusion using (SVM-EPF-G-g ∨ CEM-Gaussian) ∨ OTVCA-Gaussian.

as good as one-against-one method, the P_{OA} values of the four SC-SF classifiers using one-against-all were still higher than the P_{OA} values produced by fusing two SC-SF classifiers by 0.03%, 0.04%, 0.02%, and 0.25% for the four data sets, respectively.

C. Progressive Fusion of Three SS Classifiers

In this section, three SC-SF classifiers, SVM-EPF, CEM-Gaussian, and OTVCA-Gaussian, were fused progressively. Three different combinations of progressive SC-SF fusion methods, (SVM-EPF-G-g ∨ CEM-Gaussian)

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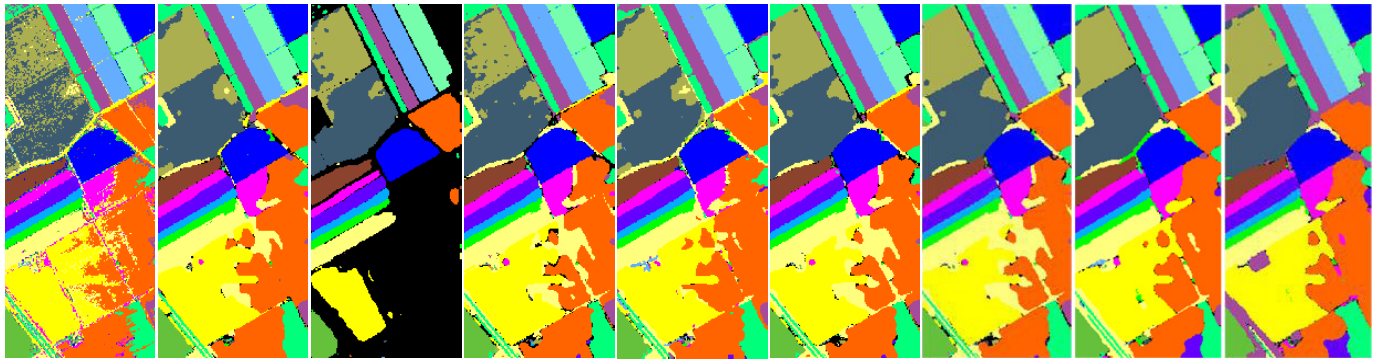


Fig. 12. Classification maps for Salinas data. (a) OTVCA. (b) ISVM-EPF-G-g. (c) ICEM-Gaussian. (d) OTVCA ∨ SVM-EPF-G-g. (e) OTVCA ∨ SVM-EPF-G-g. (f) SVM-EPF-G-g ∨ CEM-Gaussian. (g) and (h) SVM-EPF ∨ CEM-Gaussian ∨ OTVCA-Gaussian using one-against-all and one-against-one. (i) Progressive fusion using (SVM-EPF-G-g ∨ CEM-Gaussian) ∨ OTVCA-Gaussian.

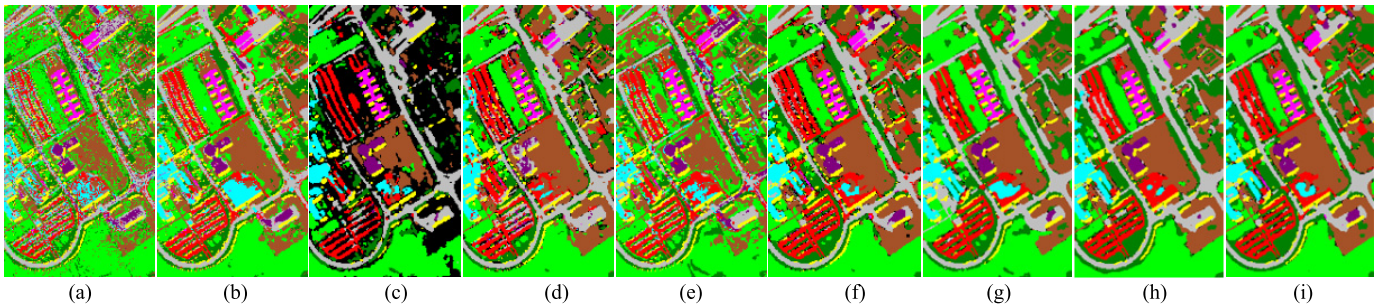


Fig. 13. Classification maps for University of Pavia data. (a) OTVCA. (b) ISVM-EPF-G-g. (c) ICEM-Gaussian. (d) OTVCA ∨ SVM-EPF-G-g. (e) OTVCA ∨ SVM-EPF-G-g. (f) SVM-EPF-G-g ∨ CEM-Gaussian. (g) and (h) SVM-EPF ∨ CEM-Gaussian ∨ OTVCA-Gaussian using one-against-all and one-against-one. (i) Progressive fusion using (SVM-EPF-G-g ∨ CEM-Gaussian) ∨ OTVCA-Gaussian.

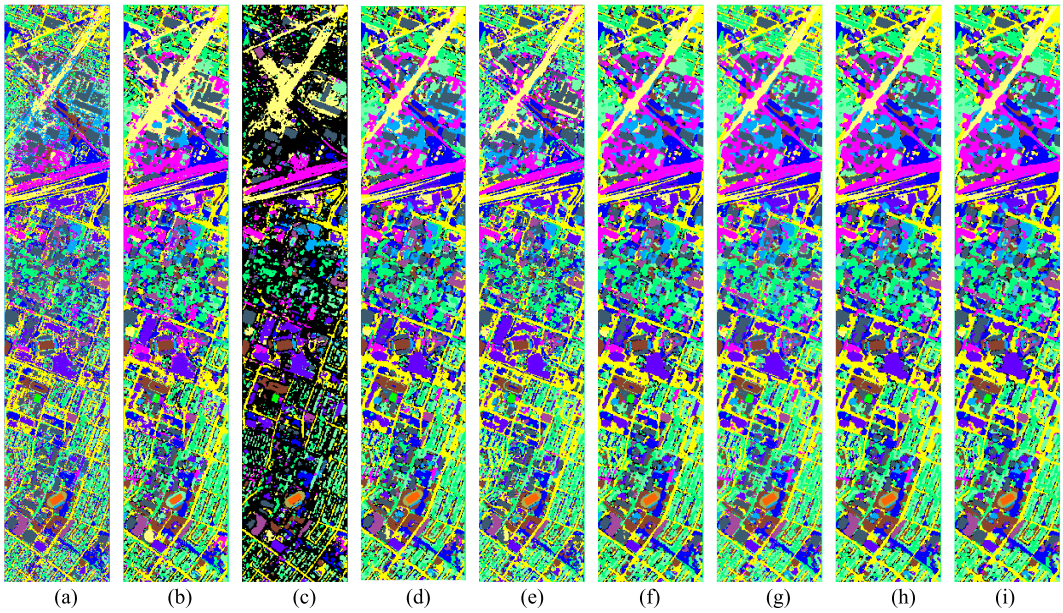


Fig. 14. Classification maps for Houston data. (a) OTVCA. (b) ISVM-EPF-G-g. (c) ICEM-Gaussian. (d) OTVCA ∨ SVM-EPF-G-g. (e) OTVCA ∨ SVM-EPF-G-g. (f) SVM-EPF-G-g ∨ CEM-Gaussian. (g) and (h) SVM-EPF ∨ CEM-Gaussian ∨ OTVCA-Gaussian using one-against-all and one-against-one. (i) Progressive fusion using (SVM-EPF-G-g ∨ CEM-Gaussian) ∨ OTVCA-Gaussian.

∨ OTVCA-Gaussian, (SVM-EPF-G-g ∨ OTVCA-Gaussian) ∨ CEM-Gaussian, (CEM-Gaussian∨ OTVCA-Gaussian) ∨ SVM-EPF-G-g, were performed for experiments with their P_A, P_{PR}, and P_{PR} (without BKG) values also tabulated in Table X with the running times t in seconds and the number of iterations, n_{iter}, listed at the last three columns in the table.

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TABLE X

P_A, P_{PR}, AND P_{PR} (WITHOUT BKG) PRODUCED BY SVM-EPF ∨ CEM-GAUSSIAN ∨ OTVCA-GAUSSIAN USING ONE-AGAINST-ALL AND ONE-AGAINST-ONE, (SVM-EPF-G-G ∨ CEM-GAUSSIAN) ∨ OTVCA-GAUSSIAN, (SVM-EPF-G-G ∨ OTVCA-GAUSSIAN) ∨ CEM-GAUSSIAN AND (CEM-GAUSSIAN ∨ OTVCA-GAUSSIAN) ∨ SVM-EPF-G-G FOR PURDUE DATA, SALINAS, UNIVERSITY OF PAVIA AND HOUSTON WITH THE ACCORDINGLY RUNNING TIME t AND NUMBERS OF ITERATION n_{iter}

Dataset		one-against-all	one-against-one	(SVM-EPF-G-g ∨ CEM-Gaussian) ∨ OTVCA-Gaussian	(SVM-EPF-G-g ∨ OTVCA-Gaussian) ∨ CEM-Gaussian	(CEM-Gaussian ∨ OTVCA-Gaussian) ∨ SVM-EPF-G-g
Purdue	P_{AA}	0.9895	**0.9897**	**0.9926**	0.9895	0.9890
	P_{APR}	0.5598	**0.5624**	**0.5592**	0.5264	0.5218
	P_{APR}(w/o BKG)	0.9921	**0.9924**	0.9925	**0.9948**	0.9898
	P_{OA}=P_{OPR}	0.9888	**0.9892**	**0.9914**	0.9901	0.9897
	t(s)	345.9	839.5	169.2	248.8	206.3
	n_{iter}	15	11+12+11=33	15	19	16
Salinas	P_{AA}	0.9925	**0.9941**	**0.9956**	0.9903	0.9912
	P_{APR}	0.5920	**0.5934**	**0.5986**	0.5867	0.5875
	P_{APR}(w/o BKG)	0.9925	**0.9944**	**0.9960**	0.9902	0.9912
	P_{OA}=P_{OPR}	0.9880	**0.9893**	**0.9914**	0.9900	0.9905
	t(s)	869.4	1364.1	503.8	685.7	669.2
	n_{iter}	8	6+6+11=23	14	12	12
University of Pavia	P_{AA}	0.9885	**0.9906**	**0.9918**	0.9824	0.9867
	P_{APR}	0.2361	**0.2518**	**0.2562**	0.2475	0.2500
	P_{APR}(w/o BKG)	0.9816	**0.9874**	**0.9875**	0.9831	0.9830
	P_{OA}=P_{OPR}	0.9861	**0.9870**	**0.9903**	0.9874	0.9870
	t(s)	1459.5	2506.7	1395.7	1136.5	1202.4
	n_{iter}	13	12+11+18=41	28	19	21
Houston	P_{AA}	0.9805	**0.9832**	**0.9867**	0.9825	0.9836
	P_{APR}	0.1080	**0.1086**	0.1090	0.1087	**0.1095**
	P_{APR}(w/o BKG)	0.9768	**0.9830**	0.9842	0.9835	**0.9849**
	P_{OA}=P_{OPR}	0.9788	**0.9805**	**0.9831**	0.9811	0.9823
	t(s)	8114.9	15492.0	8792.5	8039.4	8159.8
	n_{iter}	8	10+9+13=32	19	16	18

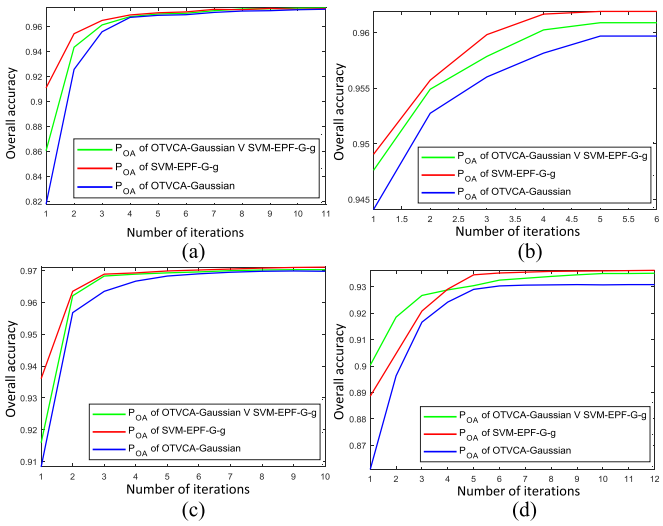


Fig. 15. P_{OA} versus iteration of OTVCA-Gaussian, SVM-EPF-G-g and OTVCA-Gaussian ∨ SVM-EPF-G-g for four data sets. (a) Purdue data. (b) Salinas. (c) University of Pavia. (d) Houston data.

According to Table X PMSSF methods produced the highest P_{OA} values among all fusion methods proposed in this article. Compared to Tables II–IX, P_{OA} values were further improved by 0.29%, 0.38% 0.44% and 0.68% respectively. Among all three combinations of progressive fusion methods, (SVM-EPF-G-g ∨ CEM-Gaussian) ∨ OTVCA-Gaussian performed the best on all the four data sets.

D. Discussion

Several interesting findings can be observed from the experimental results.

1) OTVCA is generally not competitive compared to ISVM-G-g and ICEM-Gaussian. The main reason is that OTVCA is a feature extraction-based method which is a feature-level decision maker and not a pixel-level decision maker. As a result, the extracted features used for classification are very likely to lose pixel-level spectral information which may be crucial in SC-SF classification.
2) In order to make OTVCA more competitive, a new SC-SF version of OTVCA is further developed using a similar idea proposed in [27]. It is called OTVCA-Gaussian which implements OTVCA in conjunction with Gaussian filters. As a result, OTVCA-Gaussian indeed performs better than OTVCA implemented alone because including spatial Gaussian filters can capture spatial information among extracted features. This also explains why OTVCA-Gaussian generally did better than OTVCA in all experiments for the four data sets. Nevertheless, OTVCA-Gaussian did not perform as we expected

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since the added spatial information is not sufficiently enough to make up the loss of the pixel spectral information incurred by OTVCA. Moreover, OTVCA and OTVCA-Gaussian could be further improved by fusing other SC-SF classifiers. Unfortunately, such fusion was not good enough for them to compete against other SC-SF classifiers because the added spatial information obtained from applying Gaussian filters to OTVCA-extracted features cannot compensate the loss of pixel spectral information.

3) In simultaneous fusion of three SC-SF classifiers the one using the one-against-one strategy performed better than that using the one-against-all strategy. This is the same conclusion drawn from SVM which performed multiple-class classification using the one-against-one strategy performs better than that using the one-against-all strategy [34].
4) Among three progressive fusion methods, (SVM-EPF-G-g $\vee$ CEM-Gaussian) $\vee$ OTVCA-Gaussian, performed better than (SVM-EPF-G-g $\vee$ OTVCA-Gaussian) $\vee$ CEM-Gaussian and (CEM-Gaussian $\vee$ OTVCA-Gaussian) $\vee$ SVM-EPF-G-g. In other words, pure pixel-based classifier must be fused first and the loss of abundance fractions caused by pure pixel classifiers will be made up by a mixed classifier CEM-Gaussian so as to improve classification accuracy.
5) Progressive fusion methods generally performed the best among all fusion methods and SMSSF.
6) Generally speaking, fusing the three SC-SF classifiers, SVM-EPF, CEM-Gaussian and OTVCA-Gaussian, can achieve better results than fusing any pair of these three SC-SF classifiers. However, it does not imply that fusing more SC-SF classifiers will always do better than fusing fewer SC-SF classifiers. It really depends upon SC-SF classifiers to be fused. For example, if we fuse two pure pixel-based classifiers such as SVM-Gaussian and SVM-EPF with a mixed pixel classifier, CEM, the resulting performance would not be better than fusing SVM-EPF with CEM-Gaussian. The results are not included here due to limited space. This is because SVM-Gaussian and SVM-EPF make use of the same pure pixel-based SVM classifier and SVM-Gaussian performs worse than SVM-EPF. In this case, fusing SVM-Gaussian and SVM-EPF only degrades the performance of SVM-EPF.
7) Comparing Table X to Tables II–IX the classification performance by fusing three SC-SF classifiers was not improved much over that by fusing two SC-SF classifiers. This implied that for a classifier which already performs very well it will be very difficult to improve its performance by a very small percentage. In this case, even a slight improvement is significant.

X. Complexity Analysis of Fusion of SS Classification

Fusing two SS classifiers described in Figs. 1 and 2 is designed and developed based on the concept of an iterative process to produce a fused SS classifier which is supposed to improve an individual SS classifier. So, an iterative process is carried out by three elements, initial condition which initializes the process to get it started, learning rule which improves the process performance iteration after iteration and stopping rule. In the SS fusion algorithm proposed in this article, the initial condition is initialized by a set of randomly selected training samples, the learning rule is specified by MAP in (1) and the stopping rule is determined by TI in (3). In order to see how much improvement of P_{AA} and P_{OA} could be gained by fusion, we consider three methods described in Section VIII-A of fusing two SS classifiers, OTVCA-Gaussian $\vee$ SVM-EPF-G-g, OTVCA-Gaussian $\vee$ CEM-Gaussian and SVM-EPF-G-g $\vee$ CEM-Gaussian for the four data sets where OTVCA-Gaussian and SVM-EPF-G-g used randomly selected training samples

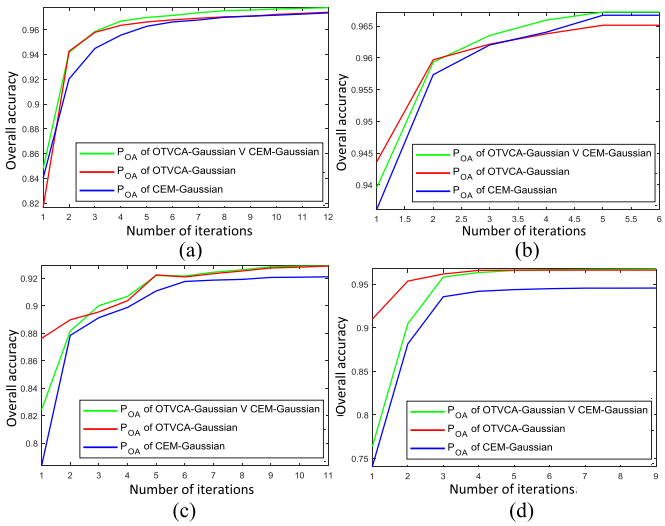


Fig. 16. P_{OA} versus iteration of OTVCA-Gaussian, CEM-Gaussian and OTVCA-Gaussian $\vee$ CEM-Gaussian for four data sets. (a) Purdue data. (b) Salinas. (c) University of Pavia. (d) Houston data.

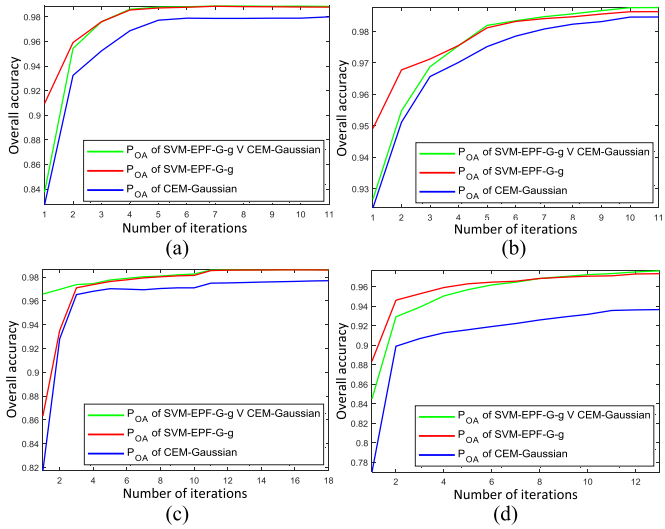


Fig. 17. P_{OA} versus iteration of SVM-EPF-G-g, ICEM-Gaussian and SVM-EPF-G-g $\vee$ CEM-Gaussian for four data sets. (a) Purdue data. (b) Salinas. (c) University of Pavia. (d) Houston data.

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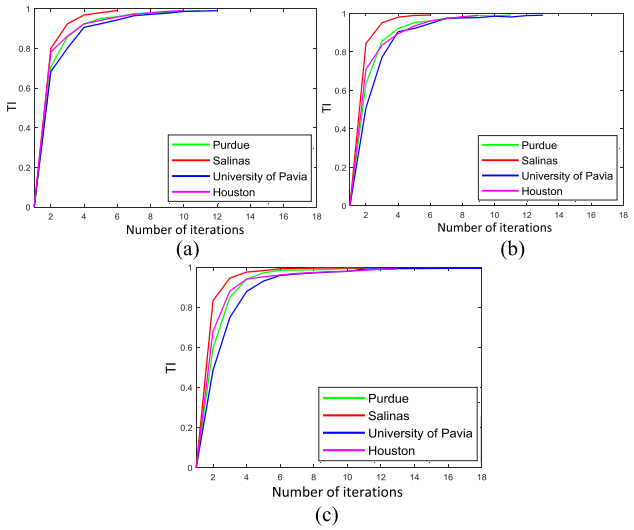


Fig. 18. TI value versus n_{iter} for Purdue, Salinas, University of Pavia and Houston data sets. (a) OTVCA-Gaussian ∨ SVM-EPF-G-g. (b) OTVCA-Gaussian ∨ CEM-Gaussian. (c) SVM-EPF-G-g ∨ CEM-Gaussian.

and CEM-Gaussian used class means. Figs. 15–17 show the values of P_{OA} produced by individual classifiers and fusion methods respectively for the four data sets. Since SVM-EPF-G-g already performed very well, its improvement by the fused algorithm is gradually increasing but the differences were not visibly appreciable. The fusion process was terminated when the improvement was stopped by the criterion either (2) or (4). On the other hand, the fused SVM-EPF-G-g∨CEM-Gaussian clearly outperformed ICEM-Gaussian all the way through even after the fusion was terminated. However, the fusion process is terminated provided that one of the algorithms to be fused, which is SVM-EPF-G-g in our case, began to be competitive to the fused algorithm.

As a final experiment, the curve plots in Fig. 18 also showed that in all three fusion methods, the value of TI versus the number of iterations, n_{iter} was increased for all the four data sets as n_{iter} was increased. It was terminated when $\text{TI}^{(k)}$ met the stopping rule (4) after 6–18 iterations. According to all our experiments, the sequences of $\text{TI}^{(k)}$ were always increasing until they reached a certain value when they began to fluctuate within a very small range of values. The stopping rule (4) simply terminates the fusion process once $\text{TI}^{(k)}$ began to fluctuate.

XI. Conclusion

This article presents a new approach to fusing multiple SS (SC-SF) classifiers. Its main contributions are summarized as follows. One is to introduce an iterative process to fuse SFMaps produced by each of SC-SF classifiers so that the resulting fusion of multiple SC-SF classifiers can take advantage of strengths of individual single SC-SF classifiers to improve classification over individual single SC-SF classifiers. Second, it makes use of MAP from detection theory to fuse multiple classifiers. Then the fused SFMaps are fed back to be added to the current hyperspectral image cube to create a new image data iteratively for the next round fusion processing. Third, to realize applicability of fusing multiple SC-SF classifiers two versions of fusing multiple SC-SF classifiers are also developed. One is SMSSF which fuses all SC-SF classifiers together simultaneously at the same time. The other is PMSSF which fuses two SC-SF classifiers at a time progressively. Fourth, to see how such fusion process works, fusion of two SC-SF classifiers is first presented and then extended to fusion of three or more SC-SF classifiers. Finally, the experimental results demonstrate that as long as SC-SF classifiers are selected properly, fusing more SC-SF classifiers generally performs better than fewer SC-SF classifiers. Moreover, compared to SMSSF PMSSF performs slightly better. In the future an investigation on fusion of heterogeneous classifiers such as fusing an SS classifier and a CNN classifier will be interesting and challenging because the structures of these classifiers are quite different and dissimilar in nature. It may require new ideas to develop an iterative version of CNN (ICNN) which is currently undertaken.

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